DSA – 20 Questions with Answers

🔹 **Short Questions (1–10)**

1. What is Data Structure?

Answer:
A data structure is a way of organizing and storing data efficiently.

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2. What is an algorithm?

Answer:
An algorithm is a step-by-step procedure to solve a problem.

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3. What is time complexity?

Answer:
Time complexity measures how much time an algorithm takes to run.

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4. What is space complexity?

Answer:
Space complexity measures how much memory an algorithm uses.

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5. What is an array?

Answer:
An array is a collection of elements stored in contiguous memory locations.

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6. What is a linked list?

Answer:
A linked list is a collection of nodes where each node contains data and a pointer to the next node.

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7. What is a stack?

Answer:
A stack is a linear data structure that follows LIFO (Last In First Out).

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8. What is a queue?

Answer:
A queue is a linear data structure that follows FIFO (First In First Out).

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9. What is recursion?

Answer:
Recursion is a method where a function calls itself.

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10. What is searching?

Answer:
Searching is the process of finding an element in a data structure.

🔹 Long Questions (11–20)

11. Explain types of data structures.

Answer:
Data structures are mainly divided into:

1. **Linear Data Structures**
    - Array
    - Linked List
    - Stack
    - Queue
2. **Non-Linear Data Structures**
    - Tree
    - Graph

Linear structures store data sequentially, while non-linear structures store data hierarchically.

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12. Explain array with advantages and disadvantages.

Answer:
An array stores elements of the same type in contiguous memory.

Advantages:

- Fast access using index
- Easy to use

Disadvantages:

- Fixed size
- Insertion/deletion is difficult

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13. Explain linked list in detail.

Answer:
A linked list consists of nodes where each node contains:

- Data
- Pointer to next node

Types:

- Singly linked list
- Doubly linked list
- Circular linked list

Advantages:

- Dynamic size
- Easy insertion/deletion

Disadvantages:

- Extra memory for pointers
- Slower access

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14. Explain stack with operations.

Answer:
A stack follows LIFO.

Operations:

- Push (insert)
- Pop (remove)
- Peek (top element)

Applications:

- Undo operations
- Expression evaluation

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15. Explain queue with types.

Answer:
A queue follows FIFO.

Types:

- Simple queue
- Circular queue
- Priority queue
- Deque (double-ended queue)

Used in scheduling and buffering.

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16. Explain time complexity with examples.

Answer:
Time complexity shows how execution time grows.

Examples:

- $O(1)$ → Constant
- $O(n)$ → Linear
- $O(\log n)$ → Binary search
- $O(n^2)$ → Nested loops

It helps compare algorithm efficiency.

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17. What is recursion? Explain with example.

Answer:
Recursion is when a function calls itself.

Example: Factorial
$n! = n \times (n-1)!$

It must have a base case to stop recursion.

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18. Explain searching algorithms.

Answer:
Searching is used to find elements.

Types:

- Linear Search (simple but slow)
- Binary Search (fast, requires sorted data)

Binary search is more efficient.

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19. Explain sorting algorithms.

Answer:
Sorting arranges data in order.

Types:

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort

Efficient sorting improves performance.

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20. What is tree data structure?

Answer:
A tree is a hierarchical structure.

Components:

- Root
- Nodes
- Edges

Types:

- Binary Tree
- Binary Search Tree

Used in databases and file systems.